

Transfer Nets

A *transfer net* is a Petri net extended with a set of Tr of transfer arcs of the form (p, t, q) . In a transfer net the enabling condition is exactly the same as in plain Petri Nets. However, the firing rule of transitions is changed so that when a transition t is fired:

1. First, tokens are removed from the preconditions of t .
2. Then, if there is a transfer arc $(p, t, q) \in Tr$ then all the tokens in p are transferred to q .
3. Finally, tokens are added to the post-conditions of t .

For the sake of simplicity, we assume that for each transition $t \in T$ there is at most one transfer arc in t (i.e. one arc of the form $(p, t, q) \in Tr$).

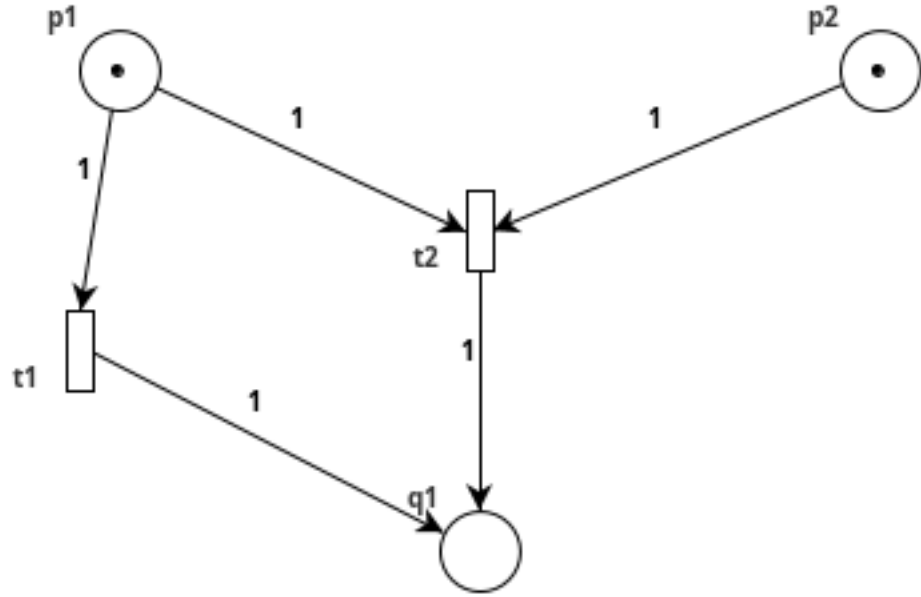
1 Formal definition of transfer nets

A *transfer net* $N = (S, T, F, Tr)$ consists of two disjoint sets of places and transitions, a set $F \subseteq (S \times T) \cup (T \times S)$ of arcs, and a set $Tr \subseteq (S \times T \times S)$ of *transfer arcs*. A transition t is *enabled* at marking M of N if $M(s) > 0$ for every place s such that $(s, t) \in F$. If t is enabled then it can *occur* leading to the marking obtained after the following operations:

1. First, remove tokens from every place s such that $(s, t) \in F$.
2. Then, remove all tokens from every place p and add them to every place q such that $(p, t, q) \in Tr$.
3. Finally, add tokens to every place s such that $(t, s) \in F$.

2 Monotonicity

In this section we will discuss the strongest form of monotonicity supported by *transfer nets*. For testing it we consider the following net:



With:

$$M_1 = \{p_1\}$$

$$M_2 = \{q_1\}$$

$$M'_1 = \{p_1, p_2\}$$

$$M'_2 = \{q_1\}$$

Although in the net we represent M'_1 instead of M_1 , the idea is clear.

Weak Monotonicity

Def. $M_1 \rightarrow M_2, M'_1 \geq M_1 \Rightarrow M'_1 \rightarrow M'_2, M'_2 \geq M_2$.

Intuitively, we know that the behaviour of these nets is the same as of plain Petri nets but some arcs are able to transfer all of their tokens to another place.

If we look at the example above, we see that $M_1 \rightarrow M_2, M'_1 \geq M_1$ and that $M'_1 \rightarrow M'_2, M'_2 \geq M_2$ so the property holds.

Strict Monotonicity

Def. $M_1 \rightarrow M_2, M'_1 > M_1 \Rightarrow M'_1 \rightarrow M'_2, M'_2 > M_2$.

Note that in case above, $M_1 > M'_1$ which is correct, but $M_2 = M'_2$ so the strict monotonicity ($>$) does not hold.

Strong Monotonicity

As we know that $StrongM \Rightarrow StrictM \Rightarrow WeakM$, and that $Strict$ doesn't hold, $Strong$ neither does.